

Bisection: Convergence

Since the error halves at each step, the method obtains an approx to within tolerance ϵ in n steps, where

$$\left. \begin{array}{l} \text{radius} \\ \text{divide} \\ \text{by 2} \\ \text{each step} \end{array} \right\} \frac{\left(\frac{b-a}{2}\right)}{2^n} < \epsilon$$

↑ solve for n .

$$\boxed{n > \log_2 \left(\frac{b-a}{2\epsilon} \right)} = O(\log_2 \left(\frac{1}{\epsilon} \right))$$

example: Find root of f in $[1, 2]$ to tolerance $\epsilon = 0.1$.

$$\text{need } n > \log_2 \left(\frac{(2-1)}{2 \cdot 0.1} \right) = \log_2 5 \approx 2.3.$$

i.e. take 3 steps.

Independent of what f is!!!

! This is really nice because now we can determine up front how many steps we need.

Recall properties of logarithms

$$\log_2(x) = \frac{\ln(x)}{\ln(2)}$$

$$\log_b(x) = \frac{\log_k x}{\log_k b}$$

$$\log(xy) = \log x + \log y$$

$$\log x^p = p \log x$$

$$\log\left(\frac{x}{y}\right) = \log x - \log y$$

$$\log \sqrt[p]{x} = \frac{1}{p} \log x$$